Exp No: 24

MODELING CLOCK-DRIVEN CONCURRENT PROCESSES USING SC_THREAD AND WAIT() IN SYSTEMC

AIM:

To understand and implement a clock-driven process using:

• sc_clock

• SC_THREAD

• wait()

• Event sensitivity (positive edge trigger)

in SystemC.

PROGRAM:

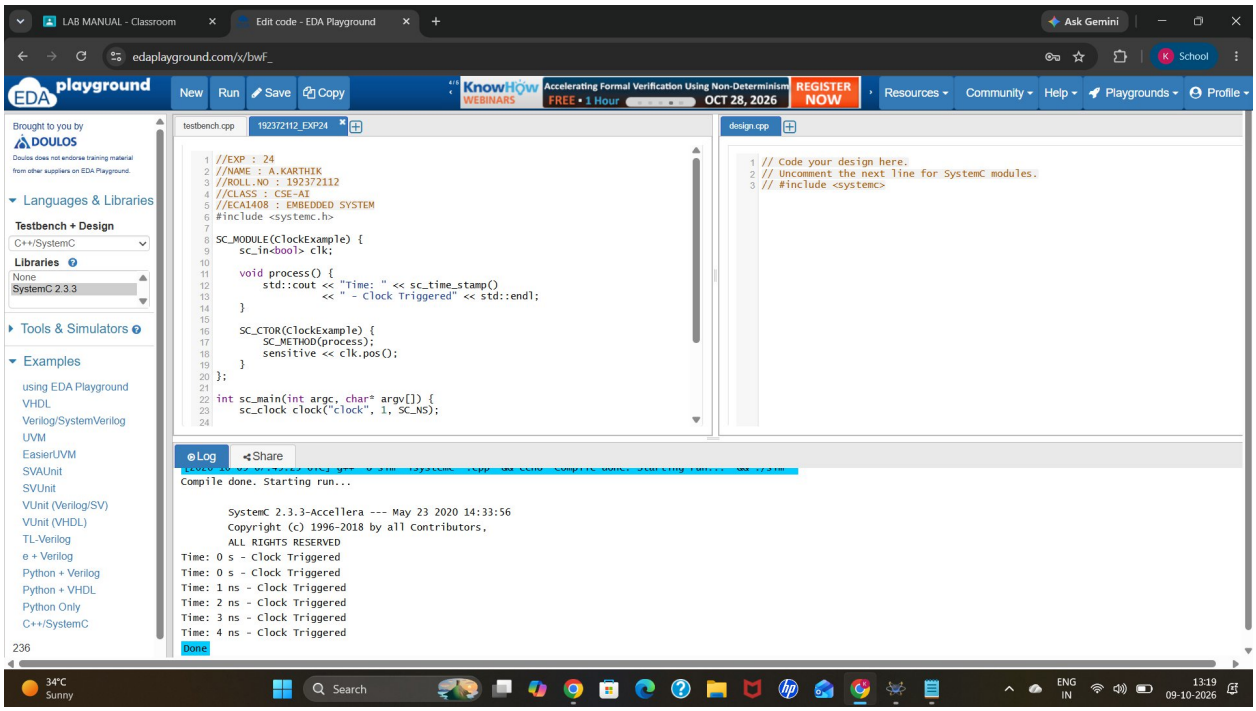
```
#include <systemc.h>
SC_MODULE(ClockExample) {
sc_in<bool> clk;
void process() {
while (true) {
wait();
std::cout << "Time: " << sc_time_stamp()
<< " - Clock Triggered" << std::endl;
}
}
SC_CTOR(ClockExample) {
SC_THREAD(process);
sensitive << clk.pos();
}
};
int sc_main(int argc, char* argv[]) {
sc_clock clock("clock", 1, SC_NS);
ClockExample module("module");
module.clk(clock);
```

sc_start(5, SC_NS);

return 0;

}

Output:



RESULT:

Thus, the clock-driven process using SC_THREAD and wait() was successfully implemented and verified in SystemC.